

Degree and Positive-Root Obstructions for Generalized Laguerre Divisibility and Quantitative Wigner Rigidity Consequences

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July 22, 2026

Status. Partial result likely valid; human review needed. The full conjecture remains open, but any counterexample outside the cases already settled in the literature is forced into a substantially smaller parameter wedge.

1 Source question

Abreu, Chabaud, Dias, and Prata study inverse problems for nodal sets of Wigner distributions [1]. Their Conjecture 2 (page 7) asks whether, for $n, k \in \mathbb{N}$ and $m \in \mathbb{N}_0$,

$$L_k \mid L_n^{(m)} \implies L_n^{(m)} = L_k, \quad (1)$$

where $L_k = L_k^{(0)}$ is the ordinary Laguerre polynomial. The paper reduces its Hermite–Wigner nodal-rigidity conjecture to (1).

We are actually able to prove the previous conjecture (see section 6) as long as we can prove the next conjecture concerning the zeros of Laguerre polynomials. We prove the conjecture in several cases and summarize other cases known in the literature.

Conjecture 2. *Let $n, k \in \mathbb{N}$, and $m \in \mathbb{N}_0$. If L_k divides $L_n^{(m)}$, then $L_n^{(m)} = L_k$.*

Figure 1: Conjecture 2 in the source paper, page 7.

We use the normalization

$$L_n^{(m)}(x) = \sum_{j=0}^n (-1)^j \binom{n+m}{n-j} \frac{x^j}{j!}. \quad (2)$$

Thus $L_n^{(m)}(0) = \binom{n+m}{n}$ and the leading coefficient is $(-1)^n/n!$. It satisfies

$$x(L_n^{(m)})'' + (m+1-x)(L_n^{(m)})' + nL_n^{(m)} = 0. \quad (3)$$

2 Proof intuition

Suppose $L_n^{(m)} = L_k R$. The two factors do not satisfy unrelated equations: substituting the product into the generalized Laguerre equation and subtracting the ordinary Laguerre equation for L_k forces

$$L_k \mid 2xR' + mR.$$

The weighted derivative on the right has degree exactly $n - k$. It is too small to contain L_k when $n < 2k$. At the only borderline case $n = 2k$, both polynomials have degree k , so they must be scalar multiples. Their constant and leading coefficients then give incompatible values for that scalar. The unresolved range therefore starts only at $n = 2k + 1$.

There is a second obstruction which uses the positivity of Laguerre zeros. For $m > 0$, set $B = 2xR' + mR$. The logarithmic derivative of R shows that B has one simple positive zero before each positive zero of R and no other zeros. Since $B = L_k T$, the quotient T also has only positive zeros. The reciprocal-root sum of a polynomial is read off from its constant and linear coefficients. Computing that sum successively for $L_n^{(m)} = L_k R$ and $B = L_k T$ forces

$$n > \frac{2(m+1)^2}{m+2} k.$$

Thus the remaining range is not merely above $2k$; for large m it begins near $2mk$.

For the Wigner consequence, the source paper's Fourier-mode induction tests

$$L_1^{(N-1)}, L_2^{(N-2)}, \dots, L_{N-1}^{(1)}$$

at every zero of L_k . The new reciprocal-root bound enlarges the direct finite-degree range from $N \leq 2k + 1$ to $N \leq 1 + 8k/3$. Combining it with the factorization ranges quoted in the source paper gives a much larger explicit bound.

3 The Laguerre divisibility obstruction

Lemma 1 (Quotient identity). *Let $n, k \in \mathbb{N}$, $m \in \mathbb{N}_0$, and suppose $L_n^{(m)} = L_k R$ for a polynomial R . Then*

$$L_k \mid 2xR' + mR. \tag{4}$$

If $r = \deg R = n - k$ and $(r, m) \neq (0, 0)$, then $2xR' + mR$ is nonzero and has degree r ; for $r = m = 0$ it is zero.

Proof. Set $P = L_k$, $Q = L_n^{(m)} = PR$, and $r = n - k$. The differential equations for P and Q are

$$xP'' + (1 - x)P' + kP = 0, \quad xQ'' + (m + 1 - x)Q' + nQ = 0.$$

Expanding the second equation by the product rule and using the first gives

$$P(xR'' + (m + 1 - x)R' + rR) + P'(2xR' + mR) = 0. \tag{5}$$

The zeros of L_k are simple, so $\gcd(P, P') = 1$. Equation (5) therefore implies that P divides $2xR' + mR$.

If $a_r x^r$ is the leading term of R , the leading term of $2xR' + mR$ is $(2r + m)a_r x^r$. Since $r, m \geq 0$, this coefficient can vanish only when $r = m = 0$. In that exceptional case the polynomial is zero; otherwise it is nonzero of degree r . In all uses below with $r > 0$, it has degree exactly r . $\square$

Theorem 2 (Degree-half range). *Let $n, k \in \mathbb{N}$ and $m \in \mathbb{N}_0$. If $n \leq 2k$ and*

$$L_k \mid L_n^{(m)},$$

then $n = k$, $m = 0$, and hence $L_n^{(m)} = L_k$. Equivalently, every nontrivial counterexample to the source conjecture must satisfy $n \geq 2k + 1$.

Proof. The case $n < k$ is impossible by degrees. Write $L_n^{(m)} = L_k R$ and $r = n - k$.

If $r = 0$, Lemma 1 says that L_k divides the constant mR . Hence $m = 0$. The common leading coefficient $(-1)^k/k!$ then gives $L_k^{(0)} = L_k$.

If $0 < r < k$, Lemma 1 makes the degree- k polynomial L_k divide a nonzero polynomial of degree $r < k$, a contradiction.

It remains to exclude $r = k$, or $n = 2k$. Put

$$2xR' + mR = cL_k.$$

The leading and constant coefficients of the factorization $L_{2k}^{(m)} = L_k R$ are

$$[x^k]R = \frac{[x^{2k}]L_{2k}^{(m)}}{[x^k]L_k} = \frac{(-1)^k k!}{(2k)!}, \quad R(0) = \binom{2k+m}{2k}.$$

Comparing leading coefficients in the displayed scalar-multiple identity gives

$$c = (2k+m) \frac{(k!)^2}{(2k)!} = \frac{2k+m}{\binom{2k}{k}}. \quad (6)$$

Comparing constant terms gives

$$c = m \binom{2k+m}{2k}. \quad (7)$$

If $m = 0$, (7) says $c = 0$, contradicting (6). If $m \geq 1$, then

$$\binom{2k+m}{2k} = \binom{2k+m}{m} \geq 2k+m \quad \text{and} \quad \binom{2k}{k} \geq 2.$$

Thus the right side of (7) is at least $2k+m$, whereas the right side of (6) is at most $(2k+m)/2$, again a contradiction. $\square$

Lemma 3 (Positive-root polar derivative). *Let $m > 0$, and let R be a degree- r polynomial with r simple positive zeros. Then*

$$B(x) = 2xR'(x) + mR(x)$$

also has r simple positive zeros. If the zeros of R are $0 < \beta_1 < \dots < \beta_r$, then B has exactly one zero in each of

$$(0, \beta_1), (\beta_1, \beta_2), \dots, (\beta_{r-1}, \beta_r).$$

Proof. Away from the zeros of R ,

$$\frac{B(x)}{R(x)} = m + 2x \sum_{j=1}^r \frac{1}{x - \beta_j}.$$

Every summand $2x/(x - \beta_j)$ has derivative $-2\beta_j/(x - \beta_j)^2 < 0$. Hence B/R is strictly decreasing on each complementary interval. It starts at $m > 0$ at the left endpoint 0, tends from $+\infty$ to $-\infty$ on each interior interval, and decreases from $+\infty$ to $m + 2r > 0$ on (β_r, ∞) . This gives the asserted r roots, all simple. Since B has degree r , there are no others. $\square$

Theorem 4 (Reciprocal-root obstruction). *Let $n, k \in \mathbb{N}$ and $m \in \mathbb{N}$. If*

$$L_k \mid L_n^{(m)},$$

then necessarily

$$n > \frac{2(m+1)^2}{m+2} k. \quad (8)$$

Proof. Write $Q = L_n^{(m)} = PR$ with $P = L_k$. Theorem 2 first gives $n \geq 2k + 1$, so R has positive degree. All zeros of Q and P are simple and positive; consequently the zeros of R are simple and positive.

For a polynomial F with positive zeros, put

$$\sigma(F) = -\frac{F'(0)}{F(0)},$$

the sum of the reciprocals of its zeros. The constant and linear coefficients in (2) give

$$\sigma(Q) = \frac{n}{m+1}, \quad \sigma(P) = k,$$

and hence

$$\sigma(R) = \frac{n}{m+1} - k. \quad (9)$$

By Lemma 1, $B = 2xR' + mR = PT$ for a polynomial T . Lemma 3 says that B has only simple positive zeros. Thus T has degree $n - 2k > 0$ and only positive zeros, so $\sigma(T) > 0$. Moreover,

$$B(0) = mR(0), \quad B'(0) = (m+2)R'(0),$$

and therefore

$$\begin{aligned} \sigma(T) &= \sigma(B) - \sigma(P) \\ &= \frac{m+2}{m} \left(\frac{n}{m+1} - k \right) - k \\ &= \frac{(m+2)n - 2(m+1)^2 k}{m(m+1)}. \end{aligned}$$

Its strict positivity is exactly (8). □

Remark 5 (Weighted critical points). If R has roots $\beta_1, \dots, \beta_r$, then away from those roots

$$\frac{2xR'(x) + mR(x)}{R(x)} = 2x \sum_{j=1}^r \frac{1}{x - \beta_j} + m.$$

Thus any hypothetical divisibility places every zero of L_k among the weighted critical points of the quotient. The degree count in Theorem 2 and the reciprocal-root argument in Theorem 4 are two consequences of this constraint.

Corollary 6 (Remaining parameter wedge). *Combine the two theorems above with the cases summarized in [1, 2]. Any counterexample not already covered by those known results must satisfy*

$$k \geq 4, \quad m > \max\{50, 5k\}, \quad n > \frac{2(m+1)^2}{m+2} k.$$

Proof. The source paper proves the conjecture for $k = 1, 2, 3$ and records that it is known for $m \leq 50$. Laishram–Shorey exclude degree- k factors when $0 \leq m \leq 5k$ and $n \geq 2k$, apart from explicit degree-one exceptions which are already disposed of in the source paper. Theorem 2 supplies the complementary range $n \leq 2k$. Theorem 4 then supplies the strict lower bound on n . □

4 Wigner nodal-rigidity consequence

We state the consequence in the finite Hermite normal form used in Section 6 of the source paper. Boundedness of the Wigner nodal set supplies such a finite normal form, and the presence of the relevant circles fixes the symplectic part as explained there.

Corollary 7 (Direct finite-degree Hermite–Wigner rigidity). *Fix $k \in \mathbb{N}$. Suppose Wf satisfies the hypotheses of Conjecture 1 of [1], and write its centered finite Hermite normal form as*

$$f = \sum_{j=0}^N c_j h_j, \quad c_N \neq 0.$$

If $N \leq 1 + 8k/3$ and

$$\mathcal{N}(Wh_k) \subseteq \mathcal{N}(Wf),$$

then $Wf = Wh_k$ everywhere.

Proof. We use the coefficient induction in Section 6 of [1]. Once $c_0, \dots, c_\ell$ have vanished, the Fourier mode $m = N - \ell - 1$ shows that $c_{\ell+1} = 0$ unless

$$L_k \mid L_{\ell+1}^{(N-\ell-1)}.$$

Put $j = \ell + 1$ and $m = N - j$. During the induction $m \geq 1$, and

$$j = N - m \leq N - 1 \leq \frac{8k}{3} = \frac{2(1+1)^2}{1+2}k \leq \frac{2(m+1)^2}{m+2}k,$$

where the last expression is increasing in $m \geq 1$. Theorem 4 therefore rules out the divisibility at every step. It follows that $c_0 = \dots = c_{N-1} = 0$.

The radial Fourier mode then says that L_k divides the ordinary Laguerre polynomial L_N . Schur's irreducibility of the ordinary Laguerre polynomials, as used in the source proof, forces $N = k$. The normalization in the source argument now gives $Wf = Wh_k$. $\square$

Corollary 8 (Literature-assisted quantitative Wigner range). *Under the same hypotheses, let*

$$M_k = \max\{50, 5k\}.$$

For $k \geq 4$, the conclusion $Wf = Wh_k$ holds whenever

$$N \leq M_k + 1 + \frac{2(M_k + 2)^2}{M_k + 3}k. \tag{10}$$

For $k = 1, 2, 3$, the source paper's proved cases give the conclusion with no restriction on the finite degree N .

Proof. As in Corollary 7, an induction step can fail only if L_k divides $L_j^{(m)}$, where $j = N - m$ and $m \geq 1$. By Corollary 6 and its proof, the divisibility is already excluded for every $m \leq M_k$. If $m \geq M_k + 1$, then

$$j = N - m \leq N - (M_k + 1) \leq \frac{2(M_k + 2)^2}{M_k + 3}k \leq \frac{2(m+1)^2}{m+2}k,$$

because $2(m+1)^2/(m+2)$ is increasing. Theorem 4 again excludes the divisibility. The final radial-mode argument is the same as in Corollary 7. $\square$

5 Verification, novelty, and limitations

The included exact-arithmetic script checks all triples

$$1 \leq k \leq 12, \quad 1 \leq n \leq 2k, \quad 0 \leq m \leq 80$$

and finds divisibility only for $n = k, m = 0$. This is a smoke test, not an ingredient of the proof.

The main proof-audit points are: (i) the sign and coefficient grouping in (5); (ii) coprimality of L_k and L'_k ; (iii) the leading and constant coefficient normalizations in the boundary case $n = 2k$; and (iv) the interlacing proof for B ; (v) the reciprocal-root normalizations; and (vi) the index shifts in the two Wigner corollaries.

Cheap run-index searches found no prior packet for arXiv:2504.20324 or this conjecture. Bounded exact and near-exact web searches on July 22, 2026 used the phrases “ L_k divides generalized Laguerre”, “common zeros Laguerre”, and “degree $2k$ Laguerre divisibility”. They found the source paper and the known factorization literature cited there, but no statement of the degree-half obstruction. In particular, the source paper’s summary of known cases does not list either the restriction $n \leq 2k$ or the reciprocal-root bound (8).

The full conjecture remains open in the wedge of Corollary 6. A further differential elimination was tested, but the second quotient T is not known to retain the original Laguerre factor, so the positive-root step cannot simply be iterated. Further arithmetic input appears necessary.

6 Human-review recommendation

Review as a likely valid substantial partial result. A reviewer should first check Lemma 1 and the two scalar values of c in the $n = 2k$ case, then check Lemma 3 and the computation of $\sigma(T)$. Finally compare Corollaries 7 and 8 line by line with the coefficient induction in Section 6 of the source paper.

References

- [1] L. D. Abreu, U. Chabaud, N. Costa Dias, and J. N. Prata, *Inverse problems for the zeros of the Wigner function*, arXiv:2504.20324, 2025.
- [2] S. Laishram and T. N. Shorey, *Extensions of Schur’s irreducibility results*, *Indagationes Mathematicae* 21 (2011), 87–105.